

Parathyroid Disorders II: Hypoparathyroidism & Approach to Hypocalcaemia

Internal Medicine Department | Source: 8-Hypoparathyroidism-
Approach_to_a_case_of_calcium_disorders.pdf (28 slides)

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 Bridge from Parathyroid Disorders I (Hyperparathyroidism):

- Same physiology: PTH raises Ca and lowers PO_4 ; corrected Ca = $Ca + 0.8 \times (4 - \text{albumin})$; **low albumin gives low total Ca without tetany**.
- This lecture is the mirror image. **Excess PTH** gave stones, bones, groans and a **short QT**. **Too little PTH** (or too little ionized Ca) gives **tetany** and a **long QT**.
- Two links to remember:
 - **Hungry bone syndrome** after parathyroidectomy was the bridge case.
 - **Thyroidectomy** (Thyroid 2 lecture) is the commonest cause of hypoparathyroidism.

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Pathophysiology and etiology of parathyroid disorders	Hereditary vs acquired causes
Knowledge	★ Manifestations, complications and management of hypoparathyroidism	Tetany signs; IV Ca gluconate
Knowledge	★ Causes and manifestations of hypocalcaemia	PTH-low vs PTH-high differential
Skill	Interpret findings and investigations → diagnosis	Ca / PO_4 / PTH / ALP pattern
Skill	Management plan for hypoparathyroidism	Acute vs maintenance
Skill	★ Emergency recognition (laryngospasm, seizures)	Post-thyroidectomy tetany

Lecture flow: Tetany (definition, causes: hypocalcaemia / alkalosis / hypomagnesaemia) → Clinical picture (overt vs latent tetany, chronic manifestations) → Investigation and treatment of tetany → Hypoparathyroidism (causes, CP, investigations, treatment) → Differential diagnosis (pseudohypoparathyroidism, pseudo-pseudohypoparathyroidism) → Lab tables → Cases.

2. Main Content (Lecture Order)

2.1 Tetany: Definition & Causes (Slides 4-6) ★

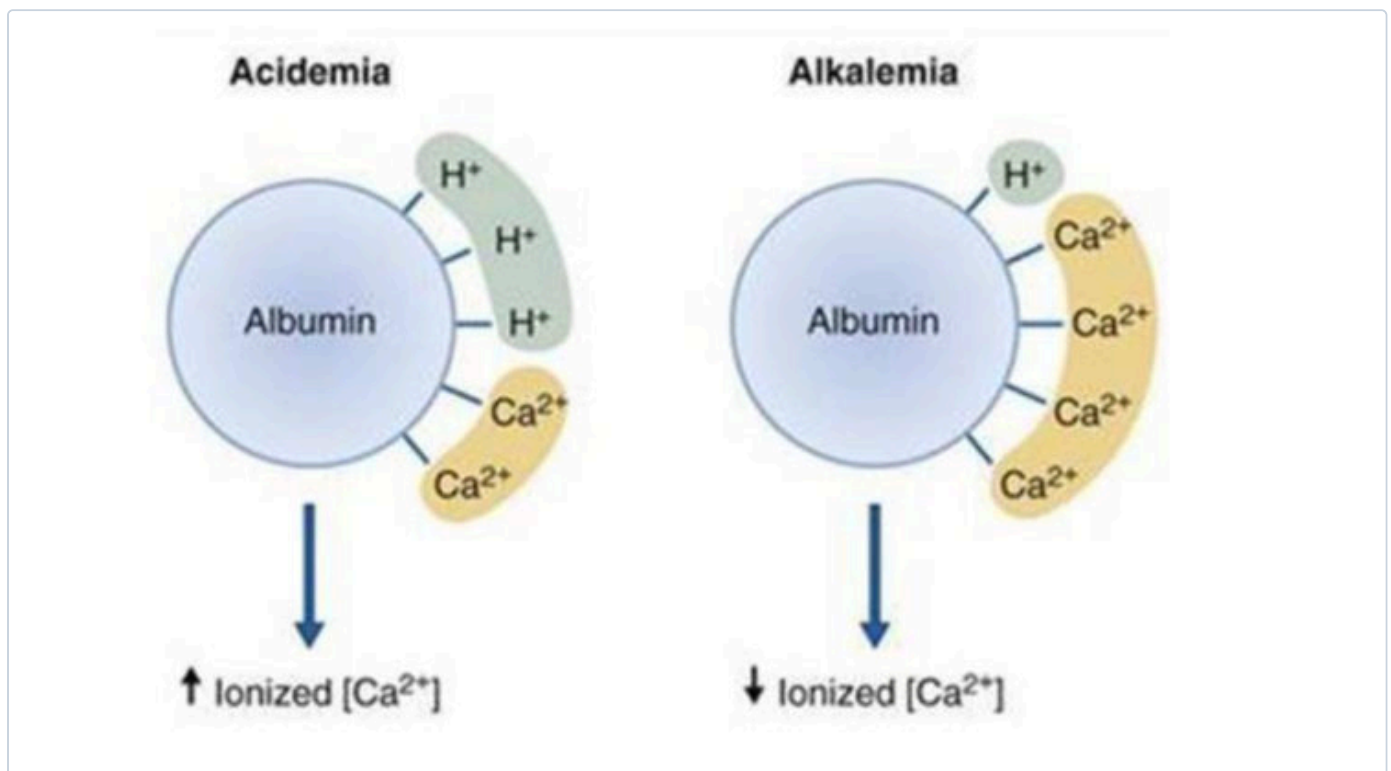
Tetany = a state of ↑ **neuromuscular excitability** with intermittent muscular spasms.

1. Hypocalcaemia: a *quantitative* ↓ of serum Ca

	Cause	Mechanism
A	★ Hypoparathyroidism	↓ PTH
B	↓ Ca intake	Prolonged starvation, anorexia, dysphagia
C	Defective absorption	Steatorrhoea (fat + Ca → Ca soaps); vit D deficiency (rickets, osteomalacia)
D	Severe acute hyperphosphataemia	Acute renal failure, tumour lysis (Ca × P precipitates)
E	Precipitation of Ca in tissues	★ Acute pancreatitis (saponification)

2. Alkalosis: a *qualitative* change

- **Total Ca is unchanged** but the **ionized fraction ↓**; **P is normal**.
- Mechanism: in alkalaemia, albumin releases H^+ and binds more Ca^{2+} . Acidaemia does the opposite.
- Causes: - Excess **alkali ingestion** - Frequent **vomiting** - **Hypokalaemic alkalosis** (e.g., **Conn's syndrome**) - ★ **Hyperventilation** (hysterical)



Slide 6: acidaemia vs alkalaemia, H^+ and Ca^{2+} competing for albumin

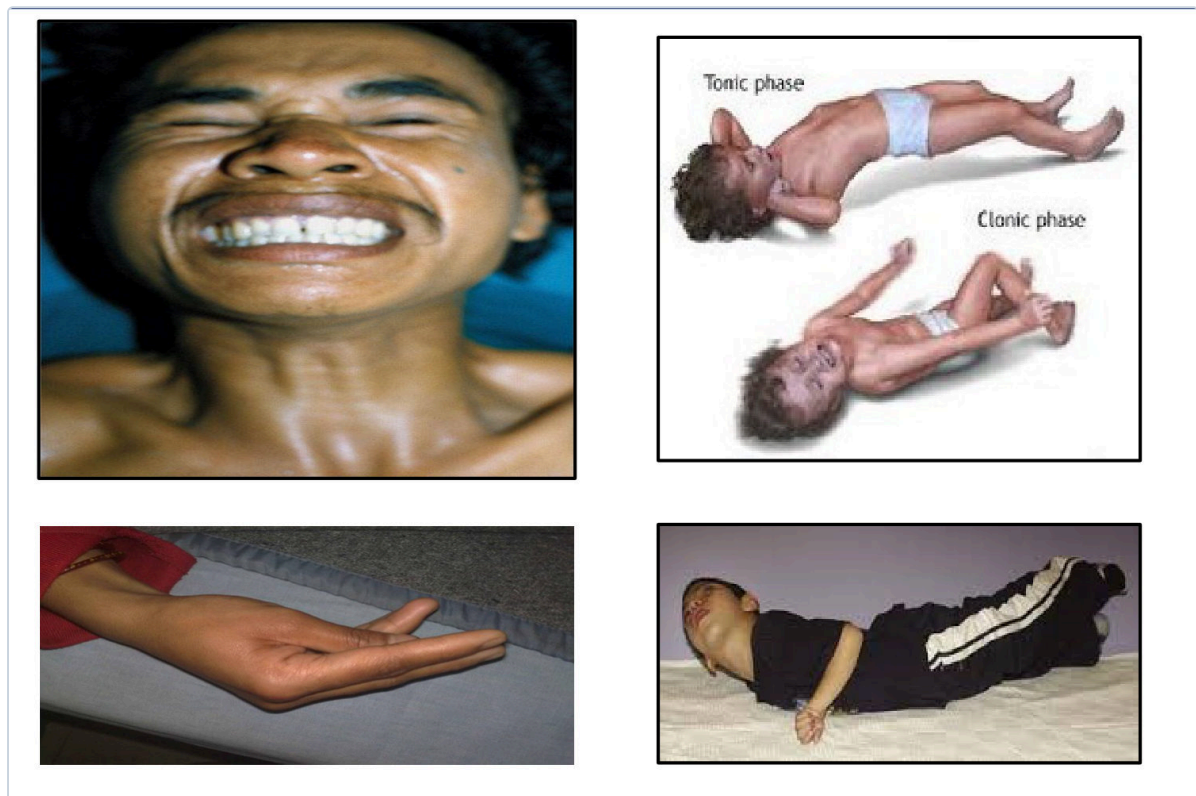
3. Magnesium tetany: ↓ Mg

- Prolonged **IV infusion without Mg supplementation**
- **Severe diarrhoea**
- **Malabsorption**
- [added] Also PPIs, alcohol, loop/thiazide diuretics, cisplatin. Low Mg **impairs PTH secretion and action**.

Mnemonic: tetany = "Low Ca, High pH, Low Mg" [added]. Only the first is a true low total calcium.

2.2 Clinical Picture: Overt (Manifest) Tetany, Serum Ca < 7 mg% (Slides 7-8) ★

	Feature	Detail
A	Paraesthesia	Extremities and around the mouth (perioral)
B	★ Carpopedal spasm	Painful spasms. Hands: adduction of the thumb, flexion of the wrist and MCP joints, extension of the fingers . Feet: plantar flexion of the toes with inversion; the patient may fall on their face
C	Spasm of other muscles	★ Laryngospasm → respiratory obstruction → sudden dyspnoea in children (laryngismus stridulus), may be fatal · facial and masseter → ★ risus sardonicus · back muscles → opisthotonus
D	Tonic and clonic spasms	May end in generalised convulsions

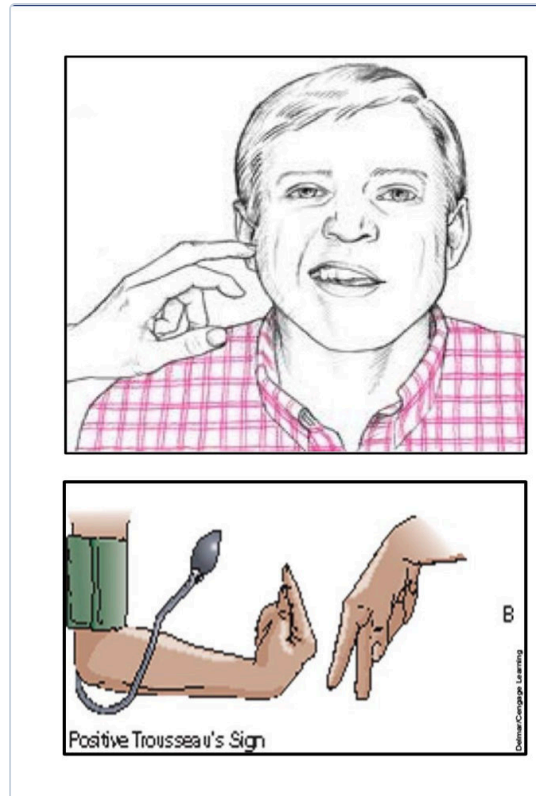


Slide 8: risus sardonicus, opisthotonus (tonic/clonic), carpal spasm

2.3 Latent Tetany, Serum Ca 7-8.5 mg% (Slides 9-10) ★

A. Provoked by:

Sign	How	Positive result
★ Chvostek's sign	Tap the facial nerve in front of the ear	Twitching of facial muscles (same side)
★ Trousseau's sign	Occlude the radial pulse with a BP cuff for 4 min	Carpal spasm (<i>main d'accoucheur</i> , "obstetrician's hand")
Erb's sign	Electrical stimulation	Contraction with < 4 mA (normally ≥ 8 mA needed)

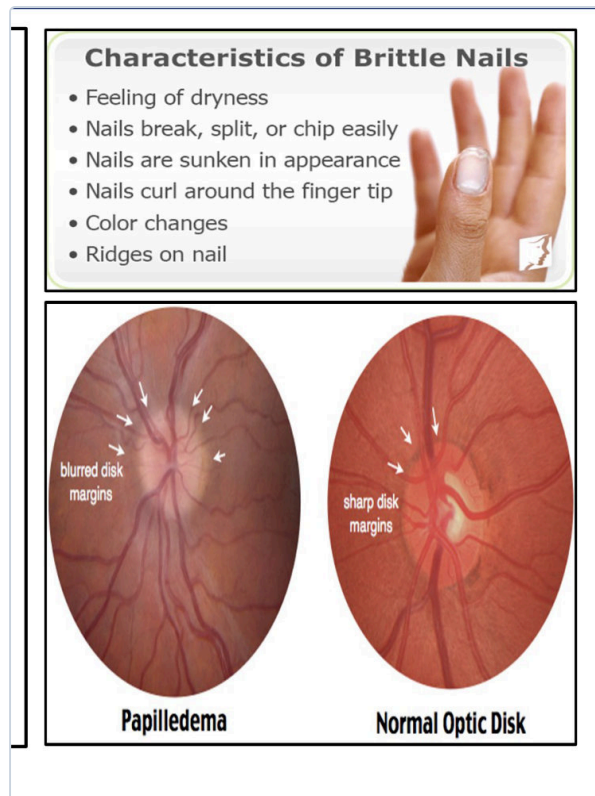


Slide 9: Chvostek's and Trousseau's signs

[added] Trousseau is more specific than Chvostek. Chvostek is positive in ~10% of normal people.

B. Manifestations (chronic):

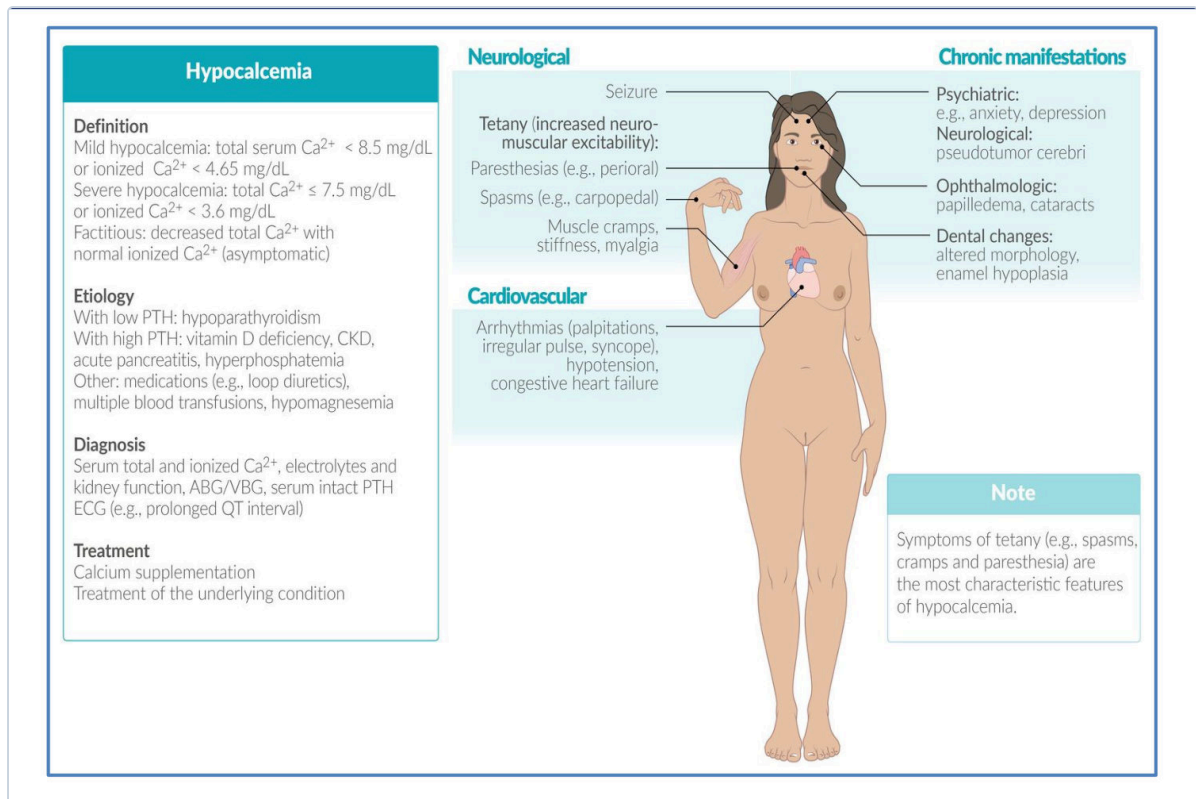
System	Features
Skin (only in hypocalcaemia , not alkalosis)	Dry skin, sparse hair, brittle nails, hypoplastic teeth
CNS	Irritability, anxiety, psychosis, epilepsy ([added] basal ganglia calcification → parkinsonism)
CVS	Hypotension, refractory heart failure, ★ prolonged QT, arrhythmias
Eyes	★ Cataract, papilloedema
Articular	Periarticular calcification (chondrocalcinosis)



Slide 10: brittle nails; papilloedema vs normal disc

Slide 20 (hypocalcaemia overview):

- **Definition:** - Mild: total Ca **< 8.5** or ionized **< 4.65 mg/dL**. - Severe: total **≤ 7.5** or ionized **< 3.6**.
- **Factitious:** low total with **normal ionized** (asymptomatic).
- **Etiology:** - Low PTH: **hypoparathyroidism**. - High PTH: **vit D deficiency, CKD, acute pancreatitis, hyperphosphataemia**. - Other: **loop diuretics, multiple blood transfusions** (citrate), **hypomagnesaemia**.
- **Diagnosis:** total and ionized Ca, electrolytes, renal function, **ABG/VBG, intact PTH, ECG (long QT)**.
- **Chronic features:** anxiety/depression, **pseudotumour cerebri**, papilloedema, cataracts, dental enamel hypoplasia.
- **Note:** **tetany (spasms, cramps, paraesthesia) is the most characteristic feature.**



Slide 20: hypocalcaemia overview

2.4 Investigation & Treatment of Tetany (Slide 11)

Investigation (for the cause):

Cause	Findings
Hypoparathyroidism	↓ PTH, ↓ serum Ca, ↑ P
Hypomagnesaemia	Low Mg; slide says " refractory hypokalaemia " ([added] also refractory hypocalcaemia : Ca won't correct until Mg is replaced)
Alkalosis	↑ pH, ↓ ionized Ca (total Ca normal)

Treatment:

- **a. Acute attack:** ★ **IV Ca gluconate 10 mL of 10%, very slowly.**
- **b. Treat the cause:** - Hypocalcaemia → **oral Ca + vit D** - Hypomagnesaemia → **oral Mg** - Alkalosis → treat the cause; **acidifying drug**

[added]:

- **Hyperventilation tetany:** rebreathe into a bag / reassure.
- **Never give IV Ca with bicarbonate in the same line** (it precipitates).
- **Calcium chloride extravasation** causes necrosis.
- **Monitor ECG**, especially if the patient is on **digoxin** (IV Ca increases toxicity).

🔄 **Recap: Tetany.** Increased neuromuscular excitability from three causes: low Ca (hypopara, low intake or absorption, high PO₄, pancreatitis), alkalosis (normal total, low ionized: vomiting, Conn's, hyperventilation), or low Mg. Overt (< 7): perioral paraesthesia, carpopedal spasm, laryngospasm, risus sardonicus, opisthotonus, fits. Latent (7–8.5): Chvostek, Trousseau (4 min), Erb (< 4 mA). Chronic: dry skin/brittle nails, cataract, papilloedema, long QT, refractory HF. Acute treatment: slow IV Ca gluconate 10%, then treat the cause.

🔗 **Link:** The prototype of PTH-deficient hypocalcaemia is hypoparathyroidism itself.

👉 **Up next: Hypoparathyroidism.**

2.5 Hypoparathyroidism: Causes (Slides 12-13) ★

Type	Causes
1. Hereditary	Isolated , or combined with other endocrine deficiencies (ovaries, adrenals...) from autoantibodies → polyglandular autoimmune deficiency syndrome ([added] APS-1: hypopara + Addison + mucocutaneous candidiasis, AIRE gene) · with absent thymus → DiGeorge syndrome ([added] 22q11 deletion; CATCH-22 : Cardiac, Abnormal facies, Thymic aplasia, Cleft palate, Hypocalcaemia)
2. Acquired	★ Accidental surgical removal during thyroidectomy (commonest) · after removal of a parathyroid adenoma (the other 3 glands were suppressed by the high Ca) → temporary · intraglandular haemorrhage in the newborn (<i>tetania neonatorum</i>) · after ¹³¹I treatment of the thyroid · haemochromatosis ([added] also Wilson's, metastatic infiltration, hypomagnesaemia)

2.6 Clinical Picture & Investigations (Slide 14)

1. **Acute hypoparathyroidism** → usually **manifest tetany**.
2. **Chronic hypoparathyroidism** → **latent tetany** + **ectodermal** + **CNS changes**.

★ **NB:** soft-tissue Ca deposition is **promoted by the hyperphosphataemia of PTH deficiency**. In **vit D deficiency**, the low Ca comes with **normal or low PO₄**, so there is no such deposition.

Investigations:

- ↓ **serum Ca**
- ↑ **serum P**
- ↓ **serum PTH**
- [added] **Normal ALP** (no osteomalacia)
- [added] **Check Mg**

2.7 Treatment (Slide 15) ★

Phase	Regimen
A. Acute hypoparathyroidism (tetany)	★ Ca gluconate 20 mL 10% IV (can be ↑ to 100 mL / 4 h)
B. Maintenance	1. Ca gluconate 1.5-3 g t.d.s. + 2. Vit D (calciferol) 50,000 IU/day (1.25 mg; 1 mg = 40,000 IU), or better dihydrotachysterol (AT 10) , OR 3. ★ 1α-vit D₃ / calcitriol 0.2-1 µg/day · 4. ★ Thiazides → ↓ urinary Ca excretion · 5. Treat the cause of alkalosis if present

[added]:

- Calcitriol (or alfacalcidol) is preferred because **without PTH, the kidney cannot 1 α -hydroxylate** vitamin D.
- Aim for **low-normal Ca** (~8-8.5) to limit **hypercalciuria → stones/nephrocalcinosis**: without PTH, renal Ca reabsorption is lost.
- **Recombinant PTH** (rhPTH 1-84 / palopegteriparatide) is used for refractory cases.

🔗 **Recap: Hypoparathyroidism.** Hereditary: autoimmune polyglandular, DiGeorge (absent thymus). Acquired: thyroidectomy (commonest), post-adenoma removal (temporary), neonatal haemorrhage, ¹³¹I, haemochromatosis. Acute → manifest tetany; chronic → latent tetany + ectodermal + CNS features. Labs ↓Ca ↑PO₄ ↓PTH, and the high PO₄ drives soft-tissue calcification. Treatment: IV Ca gluconate 10% acutely; maintenance oral Ca + vit D (calcitriol best) ± thiazide.

🔗 **Link:** The same chemistry (↓Ca, ↑PO₄) with a **high** PTH means the gland is fine but the target organ is deaf.

🔗 **Up next: Differential diagnosis, pseudohypoparathyroidism.**

2.8 Differential Diagnosis (Slides 16-19) ★

I. From other causes of hypocalcaemia and tetany (§2.1). II. From other types of hypoparathyroidism:

1. Pseudohypoparathyroidism (PHP):

- **Autosomal dominant** inheritance; ★ **peripheral (end-organ) resistance to PTH**. PTH is **normal or ↑**.
- **All physical and biochemical features of hypoparathyroidism** (↓Ca, ↑PO₄).
- **Subnormal intelligence** and **basal ganglia calcification** are more frequent.
- Additional characteristic features (★ **Albright hereditary osteodystrophy, AHO**): - **Round face** - **Short stature** - **Obesity** - **Short metacarpals** ([added] 4th/5th: the "knuckle-knuckle-dimple-dimple" sign)

2. Pseudo-pseudohypoparathyroidism (PPHP):

- An **incomplete form of PHP**; may occur in **siblings** of affected patients.
- **Only the phenotype (AHO)** is present, **without hypocalcaemia or hyperphosphataemia**.



Slide 17: Albright hereditary osteodystrophy (short stature, obesity, short metacarpals)

Slide 18: molecular table

Condition	AHO	PTH	Ca	PO ₄	Urinary Ca*	Molecular defect
Hypoparathyroidism	No	Low	Low	High	High	Variable
PHP type 1A	Yes	High	Low	High	Low	GNAS defect on the MATERNAL allele
PHP type 1B	Mild	High	Low	High	Low	Imprinting defect of the maternal GNAS allele
PHP type 2	No	High	Low	High	Low	Nil (unknown)
★ Pseudo-PHP	Yes	Normal	Normal	Normal	Normal	GNAS defect on the PATERNAL allele

*Urinary Ca is "high" in hypoparathyroidism relative to the serum level, because without PTH the kidney cannot reabsorb Ca.

Condition		AHO	PTH levels	Calcium	Phosphates	Urinary Calcium*	Molecular Defect
Hypoparathyroidism		NO	Low	Low	High	High	Variable
Pseudohypoparathyroidism	Type 1A	YES	High	Low	High	Low	GNAS defect on the maternal allele
	Type 1B	Mild	High	Low	High	Low	Imprinting defect on maternal GNAS allele
	Type 2	NO	High	Low	High	Low	NIL
Pseudopseudohypoparathyroidism		YES	Normal	Normal	Normal	Normal	GNAS defect on paternal allele

Slide 18: PHP subtypes table

Mnemonic:

- **PHP = "Pseudo = PTH present but ignored"**: high PTH, low Ca.
- **PPHP = "Pseudo twice = looks only"**: AHO body, normal labs.
- **Mum's gene** gives the full disease (1A); **Dad's gene** gives the look only (PPHP). *[added]*
- **Type 1A vs type 2**: in **1A**, **PTH injection does not raise urinary cAMP** (receptor/Gsα defect). In **2**, **cAMP rises but PO₄ excretion doesn't** (post-cAMP defect). *[added]*

Slide 19: lab differential of common causes of hypocalcaemia ★

Serum	Hypoparathyroidism	Vitamin D deficiency	Pseudohypoparathyroidism
PTH	↓	↑	Normal or ↑
Calcium	↓	↓	↓
Phosphate	↑	↓	↑

Laboratory differential of common causes of Hypocalcemia

Serum levels	Hypoparathyroidism	Vitamin D deficiency	Pseudohypoparathyroidism
Serum PTH	↓	↑	Normal or ↑
Serum Calcium	↓	↓	↓
Serum Phosphate	↑	↓	↑

Slide 19: lab differential of hypocalcaemia

Abnormal calcium: correct for albumin, then let PTH decide

Corrected Ca = $\text{Ca} + 0.8 \times (4 - \text{albumin g/dL})$
or measure ionized Ca (normal total 8.5-10.5 mg/dL)

HYPERcalcaemia (>10.5)

PTH ↑ or "normal"

PTH-driven

- 1ry HPT (adenoma 90%)
- 3ry HPT (after CKD)
- MEN 1 / 2A
- Lithium, FHH *

PO₄ ↓ (unless CKD)

Not suppressed by steroids (Dent test)

PTH ↓ (suppressed)

- Malignancy: PTHrP, osteolytic mets, myeloma
- Vit D: toxicity, lymphoma, sarcoid/granuloma
- Milk-alkali, vit A
- Thyrotoxicosis, Addison, acromegaly
- Immobilisation (Paget)
- Steroids work (vit D/myeloma)

HYPOcalcaemia (<8.5)

PTH ↓ (inappropriate)

HYPOparathyroidism

- Post-thyroidectomy
- After adenoma removal
- Autoimmune, DiGeorge
- ¹³¹I, haemochromatosis
- ↓ Mg (PTH release fails)

PO₄ ↑

ALP normal

PTH ↑ (appropriate)

- Vit D deficiency / malabsorption → PO₄ ↓, ALP ↑ (osteomalacia)
- CKD → PO₄ ↑ (2ry HPT)
- Pseudohypoparathyroidism → PO₄ ↑, AHO phenotype
- Pancreatitis, tumour lysis, rhabdo (↑ PO₄), transfusion
- Loop diuretics

Picture: "stones, bones, groans, moans, thrones"

Short QT · polyuria (Ca diabetes) · dehydration
Constipation, PUD, pancreatitis · confusion → coma

Crisis (>13-14 mg/dL):

1. IV 0.9% saline 4-6 L/day (FIRST)
2. Furosemide only AFTER rehydration · calcitonin
3. IV bisphosphonate (pamidronate) · steroids if vit D/myeloma

Picture: TETANY (↑ neuromuscular excitability)

Latent (7-8.5): Chvostek, Trousseau, Erb
Overt (<7): carpopedal spasm, laryngospasm, fits
Long QT · cataract · basal ganglia Ca · dry skin/nails

Acute: IV Ca gluconate 10% slowly

Correct Mg first if low (refractory otherwise)
Chronic: oral Ca + calcitriol (1α-vit D) ± thiazide

* FHH = familial hypocalciuric hypercalcaemia (added). Alkalosis ↓ ionized Ca with normal total Ca, so tetany without "hypocalcaemia".

Added diagram: PTH-first approach to abnormal calcium

Recap: Differential. Low Ca with **low PTH + high PO₄** = hypoparathyroidism. **High PTH + low PO₄ (+ high ALP)** = vit D deficiency. **High PTH + high PO₄ + AHO** = pseudohypoparathyroidism (GNAS, maternal). **AHO + normal labs** = pseudo-pseudo (paternal). CKD also gives high PTH + high PO₄, but with ↑ creatinine.

2.9 Cases (Slides 21-25)

Three cases, all answered in §7 (Q1–Q3).

3. High-Yield Comparison Tables

Table A: Causes of Tetany

Type	Total Ca	Ionized Ca	PO ₄	PTH	pH	Key clue
Hypoparathyroidism	↓	↓	↑	↓	N	Neck scar, autoimmune, DiGeorge
Vit D deficiency / malabsorption	↓	↓	↓	↑	N	↑ ALP, bone pain, Looser zones, steatorrhoea
Pseudohypoparathyroidism	↓	↓	↑	↑	N	AHO (round face, short 4th MC)
CKD / acute renal failure	↓	↓	↑	↑	↓	↑ creatinine
Acute pancreatitis	↓	↓	N	↑	N	Saponification, epigastric pain
Alkalosis	N	↓	N	N	↑	Vomiting, hyperventilation, Conn's
Hypomagnesaemia	↓	↓	N	↓/inappropriate	N	Diarrhoea, PPIs, alcohol; Ca refractory until Mg is given
Hypoalbuminaemia	↓	N	N	N	N	No tetany (cirrhosis, nephrotic)

Table B: Overt vs Latent Tetany

	Overt (manifest)	Latent
Serum Ca	< 7 mg%	7-8.5 mg%
Typical in	Acute hypoparathyroidism	Chronic hypoparathyroidism
Signs	Paraesthesia, carpopedal spasm , laryngospasm , risus sardonicus, opisthotonus, convulsions	Only on provocation: Chvostek , Trousseau , Erb
Associated	Emergency	Ectodermal (skin, nails, teeth), cataract, CNS, long QT

Table C: Hypercalcaemia vs Hypocalcaemia (from both lectures)

Feature	Hypercalcaemia	Hypocalcaemia
Neuromuscular	↓ excitability: weakness, hypotonia	↑ excitability: tetany
ECG	Short QT	Long QT
Mental	Depression, confusion, coma	Anxiety, psychosis, seizures
Kidney	Polyuria, stones, nephrocalcinosis	(Hypercalciuria on treatment)
Eye	Band keratopathy	Cataract , papilloedema
Skin	Dry, itchy	Dry, sparse hair, brittle nails
Emergency Rx	IV saline → furosemide, bisphosphonate	IV Ca gluconate 10% (+ Mg)

Table D: Hypoparathyroidism vs PHP vs PPHP

	Hypopara	PHP 1A	PPHP
Problem	No PTH	PTH resistance	Phenotype only
PTH	↓	↑	N
Ca / PO ₄	↓ / ↑	↓ / ↑	N / N
AHO	No	Yes	Yes
GNAS	—	Maternal	Paternal
Basal ganglia Ca, low IQ	±	More frequent	—

4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Tetany = low Ca	Alkalosis → tetany with normal total Ca (low ionized)
Low total Ca = tetany	Low albumin → low total, normal ionized → no tetany
Give Ca for hypocalcaemia	★ If Mg is low, Ca won't correct: give Mg first
Hypocalcaemia → raise PTH	Hypopara and hypo-Mg → PTH low/inappropriately normal
↓Ca + ↑PO ₄ = hypoparathyroidism	PHP and CKD also give ↓Ca ↑PO ₄ , but with high PTH
Vit D deficiency mimics hypopara	PO₄ is LOW in vit D deficiency (high in hypopara); ALP is high
Pseudo = disease	Pseudo-pseudo = phenotype only, normal biochemistry
Post-adenoma hypocalcaemia = permanent hypopara	Usually temporary suppression of the other glands, or hungry bone
Ectodermal changes in tetany	Only in true hypocalcaemia , not alkalosis (slide 10)
Vit D replacement = calciferol	In hypopara, calcitriol works better (no PTH → no renal 1α-hydroxylation) <i>[added]</i>
Thiazides cause hypercalcaemia	That property is used in hypopara to ↓ urinary Ca loss

5. Clinical Correlations

- **Day 1-4 after total thyroidectomy: perioral numbness, cramps, Trousseau positive, PTH undetectable** → **post-surgical hypoparathyroidism** → **IV Ca gluconate** (Case 2).
- **After parathyroid adenoma removal with long-standing bone disease** → **hungry bone** (low PO₄, low Mg) vs transient hypopara (low PTH).
- **Child with recurrent infections, conotruncal heart defect, cleft palate, neonatal seizures** → **DiGeorge**.
- **Hypocalcaemia + mucocutaneous candidiasis + Addison's** → **APS-1**.
- **Anxious patient overbreathing with carpal spasm** → **respiratory alkalosis** → reassure, rebreathe.
- **Vomiting patient (pyloric stenosis) with tetany** → metabolic alkalosis.
- **Alcoholic on PPIs with persistent hypocalcaemia despite IV Ca** → **hypomagnesaemia** → IV Mg.
- **Acute pancreatitis + low Ca** → saponification; a **poor prognostic** sign (Ranson) *[added]*.
- **Massive transfusion** → citrate chelates Ca → tetany and arrhythmia.

- **Short, obese child with round face and short 4th metacarpals, low Ca, high PTH → PHP 1A.** Their **parent with the same look but normal labs → PPHP.**
- **Hypocalcaemia + long QT** → risk of torsades. Correct Ca before QT-prolonging drugs.

6. Rapid Revision (Last-Day)

1. **Tetany** = ↑ neuromuscular excitability. Causes: **hypocalcaemia** (quantitative), **alkalosis** (qualitative: total N, ionized ↓, P N), **hypomagnesaemia**.
2. Hypocalcaemia causes: **hypopara**, ↓ intake, ↓ absorption (**steatorrhoea → Ca soaps, vit D deficiency**), **acute hyperphosphataemia** (ARF, tumour lysis), **acute pancreatitis**.
3. Alkalosis causes: alkali ingestion, **vomiting**, **Conn's** (hypoK alkalosis), **hyperventilation**.
4. Mg tetany: **IV fluids without Mg**, severe diarrhoea, malabsorption.
5. **Overt tetany < 7 mg%:** perioral/extremity paraesthesia, **carpopedal spasm** (thumb adducted, wrist and MCP flexed, fingers extended), **laryngospasm (laryngismus stridulus)**, **risus sardonicus**, **opisthotonus**, convulsions.
6. **Latent tetany 7-8.5 mg%:** **Chvostek** (tap facial nerve), **Trousseau** (cuff 4 min → *main d'accoucheur*), **Erb** (< 4 mA vs normal ≥ 8 mA).
7. Chronic: **dry skin, sparse hair, brittle nails, hypoplastic teeth**; anxiety, psychosis, **epilepsy**; **hypotension, refractory HF, long QT**; **cataract, papilloedema**; periarticular calcification.
8. Tetany Rx: **IV Ca gluconate 10% slowly**; oral Ca + vit D; Mg if low; treat the alkalosis.
9. **Hypopara causes:** hereditary (**autoimmune polyglandular, DiGeorge**); acquired (**thyroidectomy**, post-adenoma removal (temporary), neonatal haemorrhage, ¹³¹I, **haemochromatosis**).
10. Acute → **manifest** tetany; chronic → **latent** + ectodermal + CNS.
11. Labs: **↓Ca, ↑P, ↓PTH**. The high PO₄ drives soft-tissue calcification (not seen in vit D deficiency).
12. Acute Rx: **Ca gluconate 10% IV 20 mL** (up to 100 mL/4 h). Maintenance: **Ca 1.5-3 g tds + vit D** (calciferol 50,000 IU/day, or **calcitriol 0.2-1 µg/day**) + **thiazide**.
13. **PHP:** AD, **PTH resistance**, PTH N/↑, ↓Ca ↑PO₄, **AHO** (round face, short stature, obesity, **short metacarpals**), low IQ, basal ganglia calcification.
14. **PPHP:** AHO **without** biochemical change; siblings or relatives; **paternal GNAS**.
15. **Hypopara ↓PTH ↑PO₄ · vit D def ↑PTH ↓PO₄ · PHP ↑PTH ↑PO₄.**

7. Exam-Style Questions

Q1 (lecture case, slides 21-22). A 68-year-old man with a 45-pack-year smoking history has fatigue, cramps and hand tingling. Carpopedal spasm appears during BP measurement. ALP is normal (55 U/L), the ECG shows a prolonged QT, and his father died of lung cancer. Medications: salbutamol, ibuprofen, ramipril. What is the most likely underlying cause? a) MEN · b) Ectopic hormone production · c) Medication side effect · d) Destruction of parathyroid glands · e) Albright hereditary osteodystrophy · f) Vitamin D deficiency

Answer: d. Destruction of parathyroid glands. *[reasoning added; key not on slides]*

- He has symptomatic **hypocalcaemia** with **normal ALP**. That argues **against vit D deficiency/osteomalacia** (f), where ALP rises.
- Elimination:
 - **Ectopic hormone** (PTHrP from lung cancer) causes **hypercalcaemia** (b).
 - **MEN** causes **hyperparathyroidism** (a).
 - His drugs don't cause hypocalcaemia (c).
 - **AHO** is a lifelong phenotype that presents young (e).
- In a heavy smoker, the likely mechanism is **infiltrative destruction of the parathyroids** (e.g., metastatic lung cancer), causing acquired hypoparathyroidism.
- Caveat: the stem gives no PO₄ or PTH. The answer depends on the normal ALP and on elimination.

Q2 (lecture case, slides 23-24). Four days after total thyroidectomy, a 41-year-old man has perioral numbness, upper-limb cramps and a positive Trousseau sign. Mg 1.7 mg/dL, PO₄ 4.3, 25-OH-D 20 ng/mL, **PTH 2 pg/mL**. What is the most appropriate **initial** pharmacotherapy? a) Oral calcium citrate · b) IM calcitonin · c) IV magnesium · d) IV calcium gluconate · e) Oral vitamin D · f) SC PTH

Answer: d. IV calcium gluconate. *[reasoning added; key not on slides]*

- This is **post-surgical hypoparathyroidism** (PTH almost undetectable) with **symptomatic tetany**.
- The acute treatment is **IV Ca gluconate 10%** (slides 11, 15).
- Why the others are wrong:
 - **Mg is normal** (1.7), so c is not first.
 - **Oral Ca and vit D** (a, e) are **maintenance** after the acute phase. Calcitriol is preferred because PTH is absent.
 - **Calcitonin** lowers Ca (b).
 - **PTH injection** is reserved for refractory chronic disease (f).

Q3 (lecture Case 1, slide 25). A 63-year-old smoker has 6 weeks of cramps and perioral/acral tingling, and a positive Chvostek sign. ALP normal, **PO₄ 5.2 mg/dL** (high), creatinine 1.1 (normal), long QT. What is the most likely cause? a) Vitamin D deficiency · b) Ectopic hormone production · c) Destruction of parathyroid glands · d) 1ry hyperparathyroidism

Answer: c. *[reasoning added]*

- **Hypocalcaemia + hyperphosphataemia + normal renal function + normal ALP = hypoparathyroidism** pattern (slide 19).
- Vit D deficiency gives **low** PO₄ and high ALP.
- Ectopic hormone and 1ry HPT cause **hypercalcaemia**.

Q4. A patient has low Ca, high PO₄ and high PTH. They are short and obese, with a round face and short 4th and 5th metacarpals. Their mother has the same body habitus but normal calcium. What does the patient have? A. Hypoparathyroidism · B. Vitamin D deficiency · C.

Pseudohypoparathyroidism type 1A · D. Pseudo-pseudohypoparathyroidism · E. CKD

Answer: C.

- **End-organ PTH resistance** (high PTH, low Ca, high PO₄) **with AHO** = PHP 1A (**maternal GNAS**).
- **D** describes the phenotype with **normal** biochemistry.
- *[added]* This is the classic pedigree. The mother has **PPHP**: she inherited the GNAS mutation from her father, so she has the look but normal labs. She passes it on **maternally**, so her child gets **PHP 1A**.

Q5. A 30-year-old woman has a panic attack with tingling and carpal spasm. Total Ca and albumin are normal; ABG shows pH 7.55, PaCO₂ 24. What is the mechanism and management? A.

Hypoparathyroidism; IV Ca · B. **Respiratory alkalosis → ↓ ionized Ca; rebreathing/reassurance** ·

C. Hypomagnesaemia; IV Mg · D. Vit D deficiency; calciferol · E. Hypoalbuminaemia; no treatment

Answer: B.

- **Alkalosis** increases Ca binding to albumin → **↓ ionized Ca** with **normal total Ca** (slide 6).
- Treat the cause (slide 11).

8. Exam Pitfalls & Traps

- **Dose conflict in the lecture:** slide 11 says acute tetany → **Ca gluconate 10 mL 10% IV very slowly**. Slide 15 says **20 mL 10% IV, can be increased to 100 mL / 4 h**. Both are 10% calcium gluconate given slowly. *[added]* Common practice: 10-20 mL over 10 min, then infusion if needed.
- **Slide 11 wording:** "hypomagnesaemia → **refractory hypokalaemia**". Hypomagnesaemia also causes **refractory hypocalcaemia**, which is the point in tetany questions: **replace Mg or the Ca won't stick**.
- **Alkalosis tetany has NORMAL total Ca** and normal P. Don't pick "hypocalcaemia" as the mechanism.
- **Hypoalbuminaemia** = low total Ca **without** tetany. Correct the Ca first.
- **Ectodermal changes** (skin, hair, nails, teeth) occur **only in true hypocalcaemia**, not in alkalosis.
- **PO₄ is the discriminator:** high in **hypopara, PHP, CKD**; low in **vit D deficiency** (2ry HPT kicks in).
- **ALP:** normal in hypopara; **high in vit D deficiency/osteomalacia**. The lecture's Q1 relies on this.
- **Pseudo-pseudo has NORMAL labs.** The "pseudo twice" patient only looks the part.
- **Maternal vs paternal GNAS** (slide 18): PHP 1A = maternal; PPHP = paternal.
- **Urinary Ca "High" in hypoparathyroidism** (slide 18) means relative to the low serum Ca (lost PTH-dependent reabsorption). It also explains why over-treatment causes **stones/nephrocalcinosis** and why **thiazides** are added.
- **Post-adenoma hypocalcaemia** is usually **temporary** (suppressed glands) or **hungry bone**, not permanent hypopara.
- **Chvostek alone** can be positive in normal people. **Trousseau** is more specific *[added]*.
- **Vit D units on slide 15:** "1 mg = 40,000 IU" and "50,000 IU/day (1.25 mg)" are consistent (40 IU = 1 µg).
- **Laryngospasm in children** (laryngismus stridulus) can be fatal. Overt tetany with stridor = emergency **IV Ca**.